AAV2 Capsid Mutation-Level PCovC Analysis: Results So Far

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1 Scope

This document collects the results obtained so far from the mutation-level analysis of the AAV2 capsid biodistribution data of Ogden et al. (Ogden et al., 2019). It reports what was run and what came out. It is not a manuscript. Methods are described only as far as needed to read the numbers. Where a result has a limitation that affects how it should be read, the limitation is stated next to the result, and a summary list is in Section 4.

2 Data and methods

Data. The data are the 2,007 capsid mutants (substitutions, insertions, deletions) from the Ogden et al. deep mutational scan that passed the paper's packaging and tissue-selection filters. Each mutant has a log2 selection value in six tissues (blood, heart, kidney, lung, liver, spleen). Code and data are from the authors' public repository (`github.com/churchlab/AAV_fitness_landscape`). The paper's tropism clusters are three of nine K-means clusters (run on the 6-tissue PCA space) that the authors kept as named clusters; all other mutants are labeled "other." We use those labels as given.

Numbering and structure. Positions are in VP1 numbering. Solvent accessibility (RSA) and residue-network betweenness centrality were computed from PDB 1LP3 (Xie et al., 2002) (chain A for RSA; the 60-subunit assembly for centrality). PDB residue numbers plus 137 give VP1 numbers. 1LP3 resolves only the region common to VP1, VP2 and VP3, so positions before about residue 217 have no RSA or centrality value; 519 positions have both. Membership of a position in the 2-fold, 3-fold and 5-fold symmetry-axis contacts comes from the VIPERdb contact table included in the authors' repository. 421 positions have axis membership and a per-position selection summary.

Variable regions. The VR boundaries are those in the authors' code and in this project's scripts (Table 1). VR-III is not included, as in the authors' own aggregate list.

Per-position summary. For each position, `mean_abs_selection` is the mean over mutants at that position of the mean $|\log_2 \text{selection}|$ across the six tissues. `frac_named_cluster` is the fraction of mutants at that position assigned to one of the three named clusters. Group comparisons at the position level use two-sided Mann-Whitney U tests (two groups) or Kruskal-Wallis (several groups).

Mutation features. For each mutant, a property delta is the AAindex value of the mutant amino acid minus the value of the wild-type amino acid at that position. The original analysis used four properties (hydrophobicity, residue volume, polarity, net charge). The expanded panel uses 12 AAindex scales chosen by hierarchical clustering of all 137 scales on cross-amino-acid correlation (one representative per cluster) plus net charge, 13 in total. Deltas are set to 0 when either amino acid is unavailable (for example some indel entries). Feature deltas are z-scored before modeling.

Organ-level OLS. For each tissue, an ordinary least squares model of signed log2 selection (2,007 mutants) was fit on: the 13 property deltas, 0/1 indicators for VR-I, II, IV, V, VI, VII, VIII, IX (reference: positions outside these regions), and 0/1 indicators for insertion and deletion (reference: substitution). The model was refit on 1,000 bootstrap resamples of mutants. A coefficient is called "stable" when the 2.5th to 97.5th percentile interval of the bootstrap coefficients excludes zero. Tables give the bootstrap mean.

PCovC weights. PCovC (mixing 0.5, 2 components, class-balanced logistic regression as the internal classifier, `scikit-matter` (Goscinski et al., 2023), method of Jorgensen et al. (2026)) was fit on 25 mutation features: the 13 z-scored property deltas, 0/1 indicators for each of the eight VRs and for "no VR," and 0/1 indicators for substitution, insertion and deletion (all levels kept, no reference group). The target was the four labels (other, cluster 1, 2, 3). The p_{xz} weights, which map each feature to the classifier confidence for each class, were bootstrapped over 1,000 resamples of mutants with the same stability rule.

3 Results

3.1 Tissue-tissue correlation

Spearman correlations between tissues, computed over the 2,007 mutants, are in Figure 1. Lung, spleen, liver and heart correlate with each other at 0.52 to 0.82 (lung-spleen 0.82 is the highest). Blood correlates weakly with liver (0.04), lung (0.18), spleen (0.23) and heart (0.38), and at 0.68 with kidney.

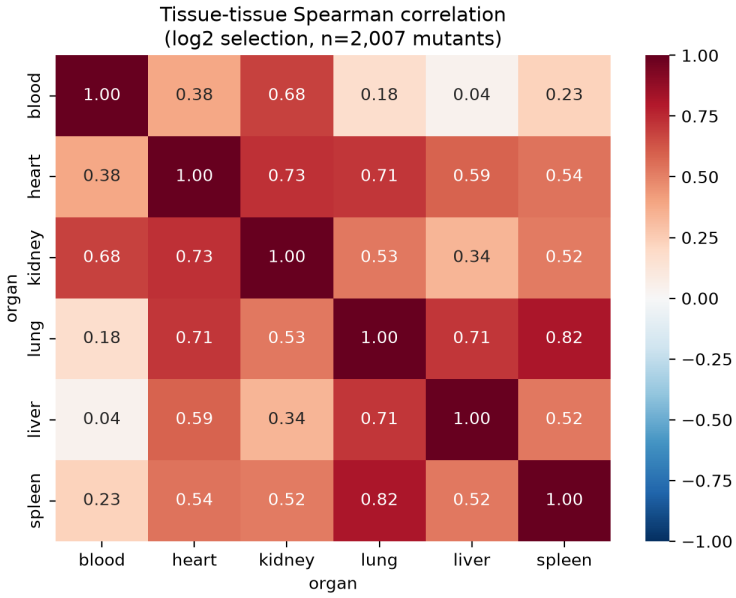


Figure 1: Spearman correlation of log2 selection between tissues, 2,007 mutants.

3.2 Symmetry axes

Positions in contact at the 3-fold axis (n=104) have higher mean absolute selection than the other positions (n=317): 1.114 vs. 0.977 (Mann-Whitney $p = 2.9 \times 10^{-6}$). The fraction of mutants in the named clusters is also higher: 21.4% vs. 12.7% ($p = 3.1 \times 10^{-6}$). Ogden et al. state that two of the

three clusters lie in surface patches near the 3-fold axis; the numbers here agree in direction with that statement.

Positions at the 5-fold axis (n=26) have lower mean absolute selection than the other positions (n=395): 0.652 vs. 1.035 ($p = 0.014$). The named-cluster fraction is 12.8% vs. 15.0% ($p = 0.125$, not significant). The 5-fold pore has been described as the route for genome packaging and for externalization of the VP1 N-terminal phospholipase domain (Bleker et al., 2005; Mietzsch et al., 2019). We have not tested whether this accounts for the lower selection at these positions.

Only four positions are on the 2-fold axis (0.587 vs. 1.015, $p = 0.270$; named-cluster fraction 50.0% vs. 14.5%, $p = 0.207$). No conclusion is drawn from that group.

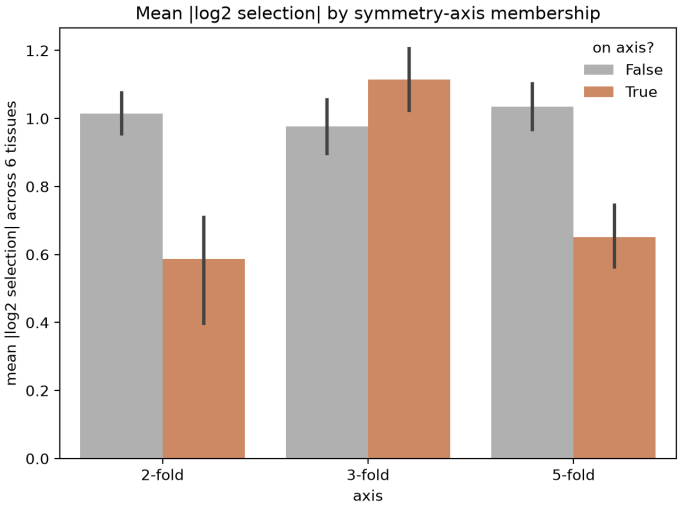


Figure 2: Mean $|\log_2$ selection$|$ across six tissues, positions on vs. off each symmetry axis. Error bars are the plotting library's default interval. n on axis: 2-fold 4, 3-fold 104, 5-fold 26; 421 positions total.

3.3 Variable regions, position level

Positions inside a VR (n=156) have higher mean absolute selection than positions outside (n=265): 1.346 vs. 0.814 (Mann-Whitney $p = 2.5 \times 10^{-21}$). Among VR positions, mean absolute selection differs by region (Kruskal-Wallis $p = 1.3 \times 10^{-16}$; Table 1). VR-VI, VR-V and VR-VIII have the highest means (1.87, 1.72, 1.61). VR-VII (0.646) is lower than the non-VR mean. VR-II has one position and VR-I and VR-IX have six each, so their means are imprecise.

Of the 156 VR positions, 42 (26.9%) are also 3-fold-axis positions, compared with 62 of 265 (23.4%) non-VR positions. Figure 3 shows mean absolute selection along the sequence with the VR intervals and the symmetry-axis positions.

Table 1: Variable-region boundaries (VP1 numbering) and per-position mean absolute selection.

Region	Residues	Positions	Mean |selection|
VR-I	262–269	6	1.126
VR-II	326–331	1	0.391
VR-IV	449–469	32	0.894
VR-V	487–504	34	1.720
VR-VI	525–540	26	1.869
VR-VII	544–556	20	0.646
VR-VIII	579–594	31	1.613
VR-IX	704–711	6	0.696
Non-VR	all other	265	0.814

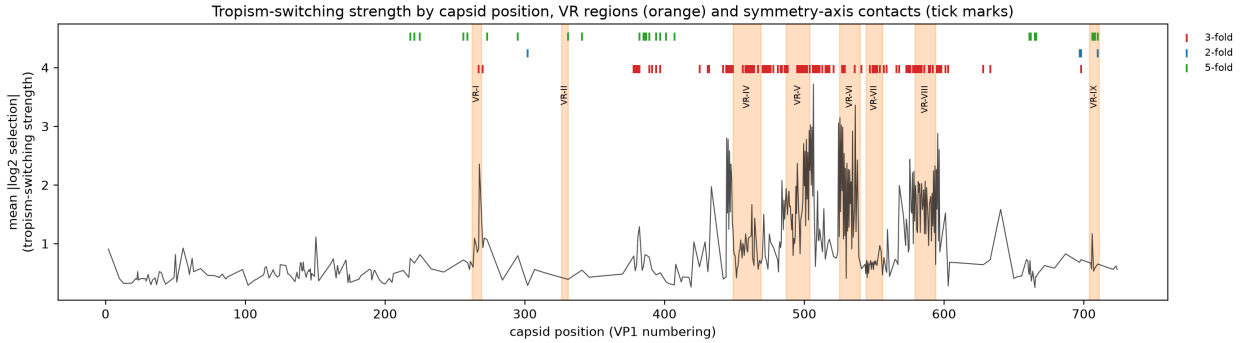


Figure 3: Mean $|\log_2$ selection$|$ by capsid position. Shaded bands are VRs. Tick marks at the top show positions in 3-fold (red), 2-fold (blue) and 5-fold (green) axis contacts.

3.4 Variable region effects by tissue

Table 2 and Figure 4 give the VR coefficients from the organ-level OLS. Each value is the difference in mean signed log2 selection for mutants inside that VR relative to mutants outside all VRs, adjusted for the other model terms. 35 of 48 coefficients are stable, compared with 20 of 78 for the property deltas in the same models.

By region: VR-VII is stable and positive in all six tissues (0.46 to 1.19). VR-IX is stable and positive in five (not liver). VR-IV is stable and positive in five (not blood). VR-V and VR-VI are stable and negative in five tissues each (not blood for either). VR-VIII is stable in all six, positive in blood (1.28), heart (0.30) and kidney (0.62) and negative in liver (-0.51), lung (-0.74) and spleen (-0.70). VR-I is stable in kidney (positive), liver (negative) and spleen (positive). VR-II (one position) has no stable coefficient.

The position-level and model-level results for VR-VII point in different directions (low mean absolute selection, Table 1, but positive signed coefficients, Table 2). The two measures differ (absolute vs. signed selection; position means vs. a mutant-level model with covariates). We have not reconciled them further.

When the bootstrap is taken over positions instead of mutants (Section 3.8), 30 of 48 VR coefficients are stable (`vr_ols_position_bootstrap_no_receptor_terms.csv`), compared with 35 of 48 in Table 2.

Insertions and deletions have negative coefficients relative to substitutions in most tissues (Table 3). The insertion coefficient is stable in heart, kidney, lung, liver and spleen. The deletion coefficient is stable in heart, lung, liver and spleen. Neither is stable in blood.

Table 2: VR indicator coefficients (bootstrap mean, log2 selection units) from the organ-level OLS. Reference group: mutants outside all VRs. Bold: 95% bootstrap interval excludes zero. Source: `expanded_organ_ols_bootstrap.csv`.

Region	Blood	Heart	Kidney	Liver	Lung	Spleen
VR-I	0.508	0.178	**0.483**	0.436	**−0.412**	**0.603**
VR-II	0.477	0.146	0.461	0.264	0.691	0.539
VR-IV	0.166	**0.464**	**0.468**	**0.613**	**0.608**	**0.191**
VR-V	0.040	**−0.226**	**−0.214**	**−0.756**	**−0.777**	**−0.530**
VR-VI	−0.040	**−0.285**	**−0.434**	**−0.898**	**−0.547**	**−1.184**
VR-VII	**0.590**	**0.458**	**0.540**	**1.194**	**0.699**	**0.914**
VR-VIII	**1.283**	**0.302**	**0.621**	**−0.745**	**−0.505**	**−0.696**
VR-IX	**0.929**	**0.621**	**0.797**	**0.663**	0.476	**0.692**

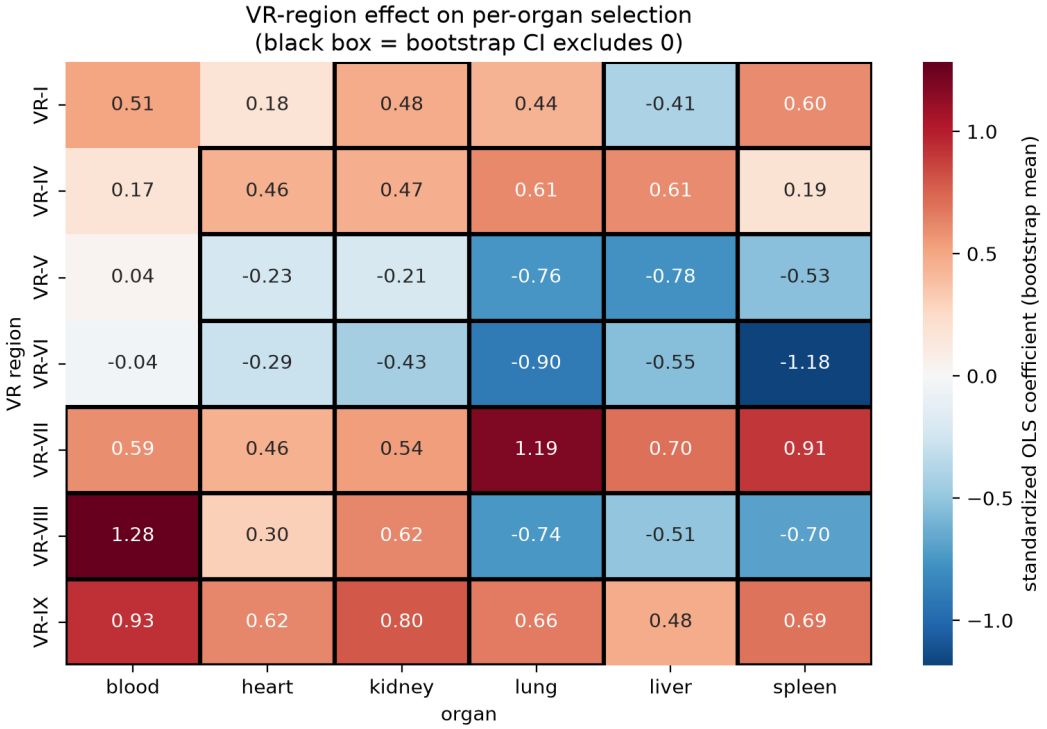


Figure 4: VR coefficients from Table 2. Black outlines mark coefficients with a bootstrap interval that excludes zero. The color bar label in the image says “standardized”; the VR and library-type indicators are 0/1 and were not rescaled. Only the property deltas were z-scored.

Table 3: Library-type coefficients (bootstrap mean) from the same models. Reference: substitution. Bold: interval excludes zero.

Type	Blood	Heart	Kidney	Liver	Lung	Spleen
Insertion	−0.048	**−0.524**	**−0.428**	**−1.096**	**−0.841**	**−0.838**
Deletion	0.236	**−0.450**	−0.205	**−0.792**	**−0.634**	**−0.555**

3.5 Physicochemical properties and tissue selection

Organ-level OLS coefficients. Of 78 property-by-tissue coefficients, 20 are stable (Table 4, Figure 5). Net charge is stable and positive in lung and spleen. Helix initiation parameter is stable and negative in lung and spleen. Apparent partial specific volume is stable and negative in heart and liver. Polarity is stable and negative in heart and liver and positive in spleen. Coil-frequency and double-bend frequency are stable and negative in blood.

Table 4: Stable property-delta coefficients from the organ-level OLS (bootstrap mean and 95% percentile interval). Coefficients are per standard deviation of the property delta.

Tissue	Property	Coefficient	95% interval
Blood	Relative frequency of double bend	−0.211	[−0.369, −0.057]
Blood	Relative frequency of coil	−0.195	[−0.286, −0.099]
Blood	AAindex “A parameter” scale	0.119	[0.040, 0.209]
Heart	Apparent partial specific volume	−0.195	[−0.303, −0.087]
Heart	Polarity	−0.102	[−0.184, −0.025]
Heart	3J(Hα-NH) coupling constant	0.068	[0.009, 0.130]
Heart	pK	0.095	[0.013, 0.179]
Kidney	Relative frequency of coil	−0.062	[−0.125, −0.001]
Lung	Helix initiation parameter	−0.168	[−0.303, −0.025]
Lung	Net charge	0.173	[0.070, 0.280]
Liver	Apparent partial specific volume	−0.296	[−0.488, −0.104]
Liver	Polarity	−0.174	[−0.316, −0.034]
Liver	Relative frequency of coil	0.089	[0.006, 0.170]
Liver	Molecular weight	0.114	[0.007, 0.222]
Spleen	Helix initiation parameter	−0.260	[−0.369, −0.153]
Spleen	pK	−0.137	[−0.249, −0.029]
Spleen	Molecular weight	−0.111	[−0.193, −0.033]
Spleen	3J(Hα-NH) coupling constant	−0.101	[−0.171, −0.037]
Spleen	Polarity	0.112	[0.013, 0.208]
Spleen	Net charge	0.178	[0.086, 0.273]

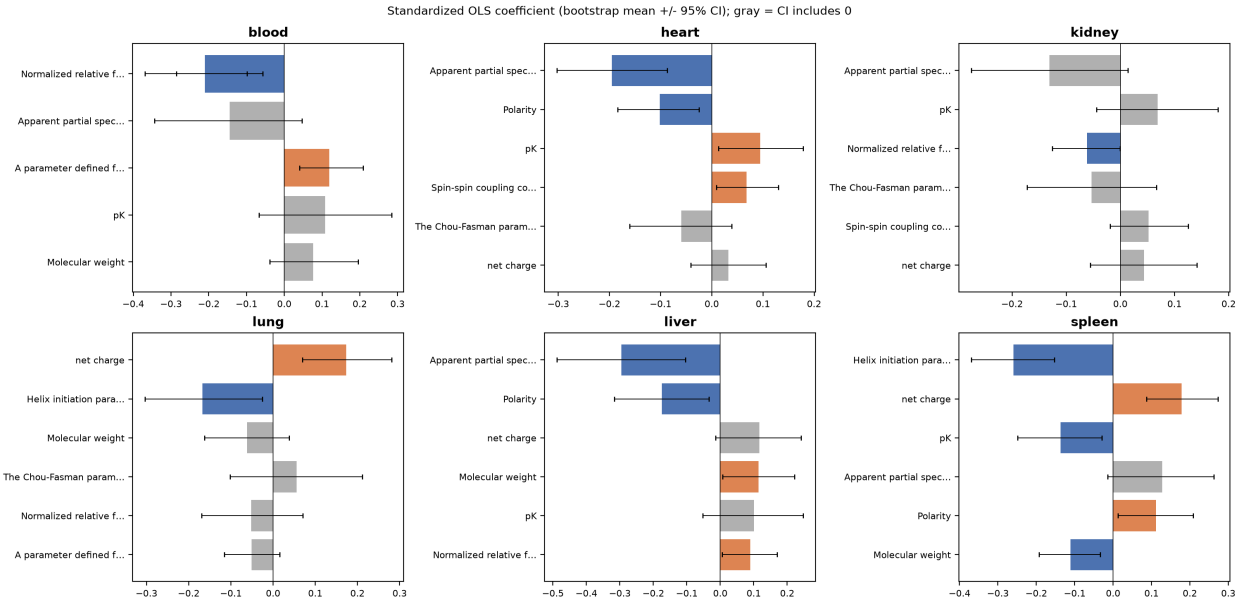


Figure 5: Top property coefficients per tissue from the organ-level OLS, with 95% bootstrap intervals. Gray bars have intervals that include zero. Only the largest coefficients per tissue are shown; Table 4 lists all stable ones.

Univariate Spearman correlations. Table 5 gives the Spearman correlation of each property delta with each tissue's log2 selection across mutants. No significance marking is shown in this table.

Table 5: Spearman correlation between property delta and log2 selection, 13-property panel. Source: `expanded_mutation_organ_spearman.csv`.

Property	Blood	Heart	Kidney	Lung	Liver	Spleen
Alpha-helix index (beta-proteins)	0.020	0.125	0.075	0.063	0.085	−0.001
Helix initiation parameter	−0.006	0.115	0.040	−0.078	0.040	−0.208
3J(Hα-NH) coupling constant	0.037	0.074	0.067	0.109	0.129	0.112
Frequency of 4th residue in turn	0.007	−0.008	−0.024	0.001	0.026	0.031
pK	0.009	0.045	0.030	−0.045	−0.050	−0.106
Molecular weight	0.022	−0.047	0.003	0.046	0.043	0.123
Chou-Fasman coil parameter	0.003	−0.034	−0.030	−0.039	−0.003	−0.041
Relative frequency of coil	0.027	0.003	−0.004	−0.037	0.001	−0.063
Apparent partial specific volume	−0.011	−0.069	−0.018	0.004	−0.085	0.040
Polarity	0.003	−0.079	−0.035	−0.073	−0.024	−0.013
Relative frequency of double bend	−0.085	−0.076	−0.075	−0.050	0.001	0.026
AAindex "A parameter" scale	0.060	−0.003	0.021	0.014	0.018	0.034
Net charge	−0.010	−0.139	−0.060	0.076	−0.062	0.205

Fit of the 4-property and 13-property models. Table 6 and Figure 6 compare the in-sample R^2 of OLS models using only the property deltas (no VR or library-type terms). The 13-property model has a higher R^2 in all six tissues, by 0.003 (lung) to 0.033 (blood). The 4-property and 13-property sets are not nested (the 4-property set includes hydrophobicity and residue volume, which are not in the 13), and the R^2 values are in-sample, not cross-validated.

Table 6: In-sample R^2, property-only OLS.

Tissue	4 properties	13 properties	Difference
Blood	0.138	0.172	+0.033
Heart	0.220	0.228	+0.008
Kidney	0.161	0.168	+0.007
Lung	0.347	0.350	+0.003
Liver	0.237	0.241	+0.004
Spleen	0.346	0.362	+0.016

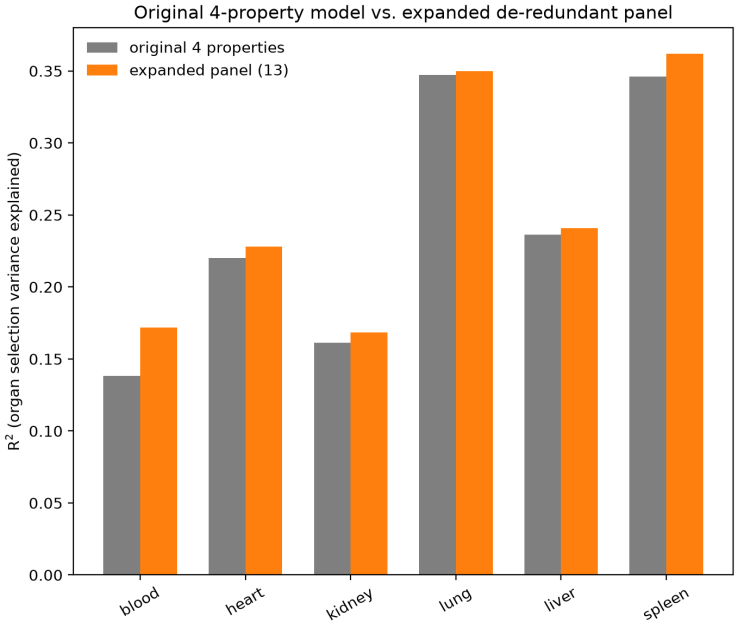


Figure 6: In-sample R^2 per tissue for the 4-property and 13-property models.

3.6 PCovC feature weights by cluster

Table 7 lists the p_{xz} weights whose bootstrap intervals exclude zero (7, 7, 5 and 5 of 25 features for the other class and clusters 1, 2 and 3, respectively). Figure 7 shows the top weights for clusters 1 to 3.

For cluster 1, net charge (negative) and apparent partial specific volume (negative) are stable, as are VR-VIII (positive) and VR-VII (negative). For clusters 2 and 3, the Chou-Fasman coil parameter and polarity are both stable with opposite signs (cluster 2: 0.061 and 0.050; cluster 3: -0.065 and -0.057). Apparent partial specific volume is stable and positive for cluster 3 (0.073) and stable and negative for cluster 1 (-0.095). The stable weights are small in absolute size (largest magnitude 0.164), and many intervals have a bound close to zero.

Table 7: Stable p_{xz} weights (bootstrap mean and 95% percentile interval).

Class	Feature	Weight	95% interval
Other	VR-VIII indicator	−0.068	[−0.120, −0.006]
	VR-VI indicator	−0.025	[−0.048, −0.002]
	VR-V indicator	−0.024	[−0.050, −0.001]
	VR-I indicator	0.009	[0.000, 0.018]
	VR-VII indicator	0.053	[0.003, 0.100]
	Apparent partial specific volume	0.074	[0.009, 0.128]
	Net charge	0.135	[0.001, 0.215]
Cluster 1	Net charge	−0.164	[−0.280, −0.004]
	Apparent partial specific volume	−0.095	[−0.180, −0.007]
	VR-VII indicator	−0.065	[−0.126, −0.005]
	VR-I indicator	−0.010	[−0.021, −0.000]
	VR-V indicator	0.029	[0.002, 0.064]
	VR-VI indicator	0.030	[0.003, 0.062]
	VR-VIII indicator	0.083	[0.008, 0.156]
Cluster 2	Substitution indicator	−0.017	[−0.041, −0.000]
	Insertion indicator	0.011	[0.001, 0.028]
	Frequency of 4th residue in turn	0.026	[0.001, 0.051]
	Polarity	0.050	[0.007, 0.096]
	Chou-Fasman coil parameter	0.061	[0.003, 0.124]
Cluster 3	Chou-Fasman coil parameter	−0.065	[−0.130, −0.002]
	Polarity	−0.057	[−0.113, −0.003]
	Insertion indicator	−0.013	[−0.029, −0.001]
	Substitution indicator	0.021	[0.002, 0.046]
	Apparent partial specific volume	0.073	[0.008, 0.144]

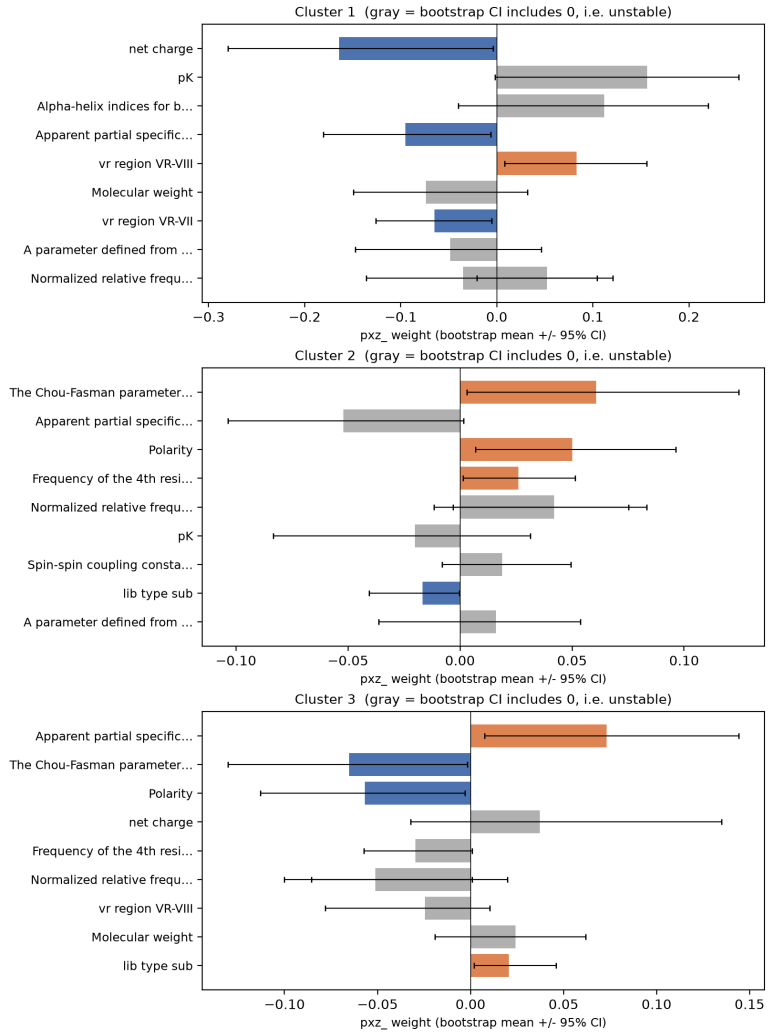


Figure 7: Top p_{xz} weights for clusters 1 to 3 with 95% bootstrap intervals. Gray bars have intervals that include zero.

3.7 Network centrality and solvent accessibility

Betweenness centrality is negatively correlated with mean absolute selection for substitutions (n=1,024, Spearman $\rho = -0.135$, 95% interval $[-0.194, -0.068]$, $p = 1.4 \times 10^{-5}$) and positively correlated for insertions (n=761, $\rho = +0.214$ [0.144, 0.286], $p = 2.3 \times 10^{-9}$). For deletions (n=37), $\rho = +0.297$ $[-0.058, 0.553]$, $p = 0.075$. Per-tissue correlations are in Table 8. A pooled model with a centrality-by-library-type interaction gives a centrality slope of -0.044 $[-0.079, -0.009]$ for substitutions, with additional slopes of $+0.265$ [0.182, 0.344] for insertions and $+0.287$ [0.067, 0.497] for deletions (Figure 8). We do not have an explanation for the difference in sign between substitutions and insertions.

RSA and centrality are negatively correlated across the 519 positions that have both ($\rho = -0.295$, $p = 6.8 \times 10^{-12}$; Figure 9). RSA is negatively correlated with mean absolute selection for substitutions ($\rho = -0.254$ $[-0.310, -0.199]$) and insertions ($\rho = -0.388$ $[-0.444, -0.327]$). For deletions the estimate is similar ($\rho = -0.266$) but the interval includes zero ($[-0.569, 0.092]$, $p = 0.112$, n=37).

Table 8: Spearman correlation of betweenness centrality with mean absolute selection and with each tissue's log2 selection, by library type (n in parentheses). Bold: 95% bootstrap interval excludes zero.

Type	Mean |sel|	Blood	Heart	Kidney	Lung	Liver	Spleen
Substitution (1024)	**−0.135**	−0.004	−0.018	**0.128**	**−0.157**	**−0.191**	**−0.217**
Insertion (761)	**0.214**	**0.214**	**0.260**	**0.220**	**0.135**	0.028	**0.223**
Deletion (37)	0.297	0.182	0.242	0.225	0.250	0.302	0.142



Figure 8: Left: centrality vs. mean absolute selection by library type. Right: interaction-model coefficients; the bars are the extra slope for deletions and insertions relative to the substitution slope, with 95% bootstrap intervals.

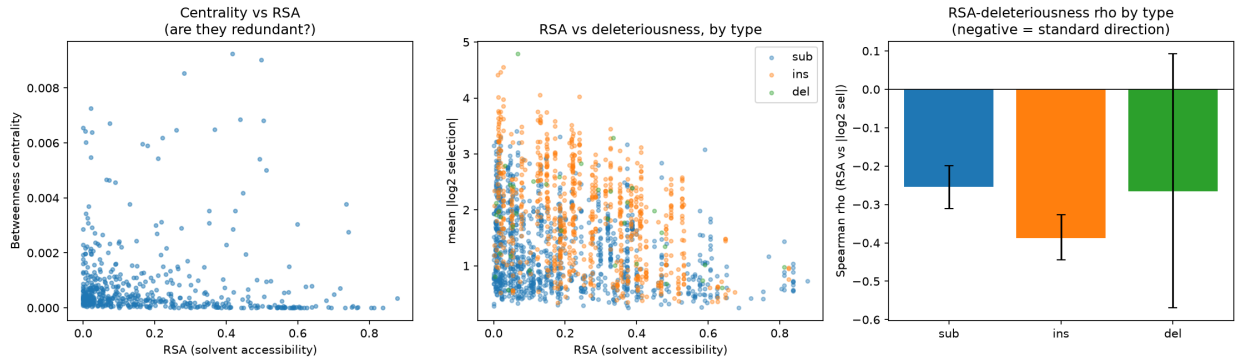


Figure 9: Left: centrality vs. RSA per position. Middle: RSA vs. mean absolute selection by library type. Right: Spearman correlation of RSA with mean absolute selection by library type, with 95% bootstrap intervals.

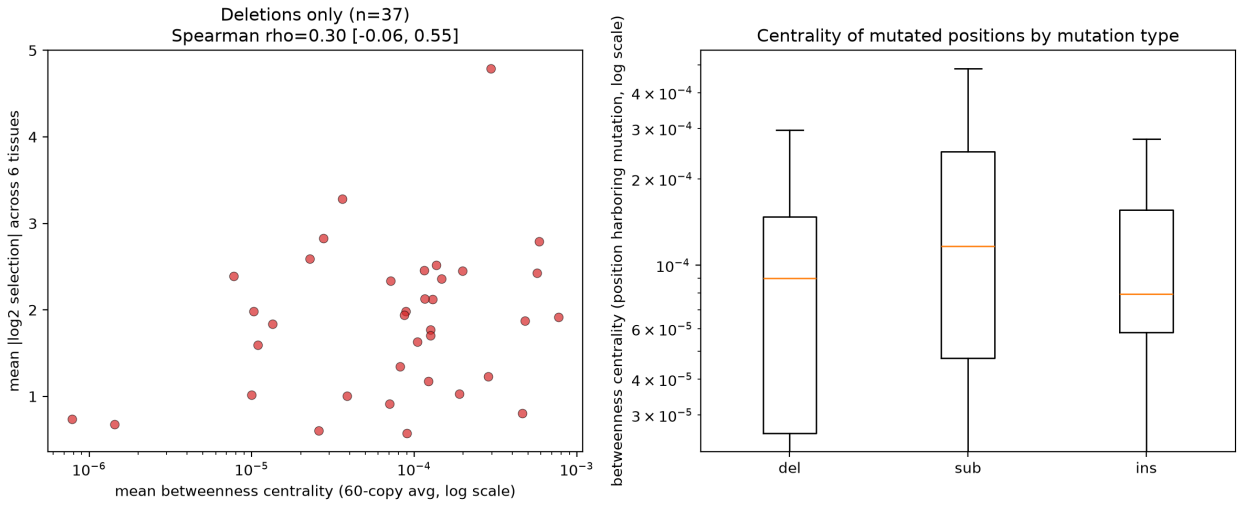


Figure 10: Left: deletion mutants only, centrality vs. mean absolute selection (n=37). Right: centrality of positions carrying each mutation type.

3.8 Receptor-associated residues

The VR results above show that mutations in some regions shift selection differently across tissues. This section asks whether specific residues that the literature assigns to receptor binding account for part of that, beyond the region they sit in.

Residue sets. Both sets are in VP1 numbering.

- *HSPG (heparan sulfate proteoglycan):* R484, R487, K532, R585, R588 (Opie et al., 2003; Kern et al., 2003). In those studies, mutation of R585 or R588 eliminated binding to heparin-agarose, and R484, R487 and K532 mutants kept partial binding. K527 is not included: Opie et al. report that it does not appear to be involved in heparin binding. The text of the AAV2-AAVR structure paper names "Arg585 and Arg587" in this context; we use R588, following Opie et al. and Kern et al.

- *AAVR (KIAA0319L):* the 19 AAV2 residues with a non-hydrogen atom within 4.5 Å of AAVR in the AAV2-AAVR PKD2 complex (Table 2 of Meyer et al. 2019). That paper describes VR-I and VR-III as the core of the footprint. Table 9 places each residue in the VR scheme used here. VR-III is not one of the regions in our model.

Table 9: Receptor-associated residues by variable region. †No mutants for this residue in the data. Positions H271 and R471 lie just outside VR-I and VR-IV; R484 lies just outside VR-V.

Region	HSPG residues	AAVR contact residues
VR-I	none	Q263, S264, G265†, A266, S267, N268†
VR-II	none	none
VR-III (not in model)	none	N382, G383†, S384, Q385
VR-IV	none	none
VR-V	R487	W502, T503
VR-VI	K532	D528, D529
VR-VII	none	none
VR-VIII	R585, R588	Q589
VR-IX	none	K706, V708
Outside VRs	R484	H271, R471

Tests. Two comparisons were made, both against the region the residue sits in so that the result does not just restate the VR enrichment. (1) Position level: the mean over positions of |selection| and of signed selection per tissue for the receptor residues, compared with the same number of residues drawn at random from the same VR strata (20,000 draws; residues outside all VRs form their own stratum). (2) Mutant level: the organ-level OLS from above with two added 0/1 indicators (HSPG residue, AAVR residue). Because mutants at one position are not independent, the bootstrap here resamples positions, with all mutants at a position kept together (1,000 draws). Mutants of every type at these integer positions are used; insertions carry half-integer position labels and are not counted. The HSPG set has 5 positions and 87 mutants. The AAVR set has 16 positions with data and 69 mutants (9 of the 16 have three or fewer mutants).

For reading the numbers: the mean log2 selection of substitution mutants outside all VRs is negative in every tissue (blood −0.90, heart −0.70, kidney −0.79, lung −1.30, liver −1.03, spleen −0.85; `non_vr_substitution_baseline.csv`), so a value near zero is better than a typical mutant.

HSPG residues. Figure 11 shows the per-residue means. R585 and R588 (20 mutants each) have the highest blood values of any residue in either set (2.7 for both), positive values in heart (0.5, 0.6) and kidney (0.5, 0.5), and negative values in lung, liver and spleen (−1.6 to −2.6). R487 and K532 are also positive in blood (1.5, 1.1). R484 (10 mutants, outside all VRs) is 0.8 in blood and has the lowest lung value (−3.5); its spleen value is −3.1.

Against residues drawn from the same VRs (Table 10), the HSPG residues have higher mean absolute selection (1.73 vs. 1.27, $p = 0.011$), higher blood selection (1.74 vs. −0.23, $p < 0.001$), higher heart selection (−0.04 vs. −0.69, $p = 0.005$) and lower spleen selection (−2.88 vs. −1.41, $p < 0.001$). Lung is lower ($p = 0.059$). Kidney and liver do not differ from region-matched residues. In the OLS (Table 11, Figure 12), the HSPG indicator is stable in blood (+2.62), heart (+0.77) and spleen (−0.80). Kidney is +0.57 with an interval that includes zero ([−0.24, 1.19]); lung and liver are near zero with wide intervals.

Liver: R585 and R588 are negative in liver (−1.8, −1.9), but so are the other residues in these regions, and the HSPG indicator for liver is −0.08 [−1.01, 0.76]. In this data the HSPG residues are not distinguishable from their region-mates in liver.

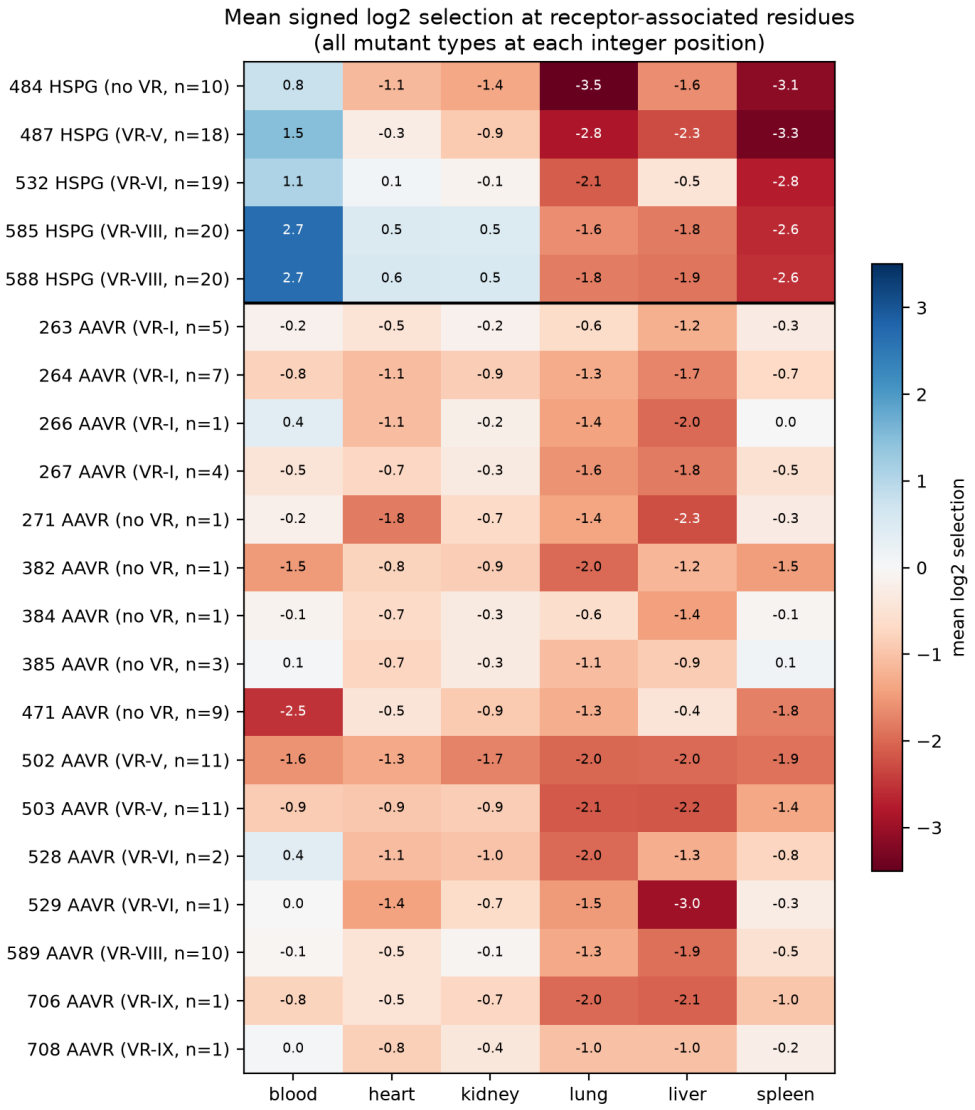


Figure 11: Mean signed log2 selection per tissue at each receptor-associated residue. Rows above the black line are HSPG residues; rows below are AAVR contact residues. Labels give the VR and the number of mutants.

Table 10: Position-level comparison of receptor residues with residues drawn at random from the same VR strata (20,000 draws). "VR-matched" is the mean of the draws. p is two-sided from the draws.

Set	Measure	Observed	VR-matched	Difference	p
HSPG (5 positions)	Mean |selection|	1.73	1.27	0.46	0.011
	Blood	1.74	−0.23	1.97	<0.001
	Heart	−0.04	−0.68	0.64	0.005
	Kidney	−0.27	−0.66	0.39	0.118
	Lung	−2.38	−1.78	−0.60	0.059
	Liver	−1.62	−1.41	−0.21	0.563
	Spleen	−2.88	−1.41	−1.47	<0.001
AAVR (16 positions)	Mean |selection|	1.04	0.98	0.05	0.513
	Blood	−0.51	−0.37	−0.14	0.478
	Heart	−0.90	−0.70	−0.19	0.057
	Kidney	−0.63	−0.58	−0.05	0.665
	Lung	−1.45	−1.34	−0.11	0.435
	Liver	−1.65	−1.32	−0.33	0.057
	Spleen	−0.69	−0.77	0.09	0.501

Table 11: Receptor-residue indicator coefficients in the organ-level OLS (bootstrap mean and 95% interval, bootstrap over positions). The model also contains the 13 property deltas, the VR indicators and the library-type indicators. Bold: interval excludes zero.

Tissue	HSPG residues	AAVR residues
Blood	**2.62 [2.06, 3.14]**	−0.07 [−0.63, 0.64]
Heart	**0.77 [0.03, 1.21]**	−0.08 [−0.38, 0.15]
Kidney	0.57 [−0.24, 1.19]	−0.01 [−0.44, 0.35]
Lung	0.06 [−1.09, 0.91]	0.20 [−0.21, 0.63]
Liver	−0.08 [−1.01, 0.76]	−0.11 [−0.54, 0.33]
Spleen	**−0.80 [−1.62, −0.20]**	0.19 [−0.33, 0.78]

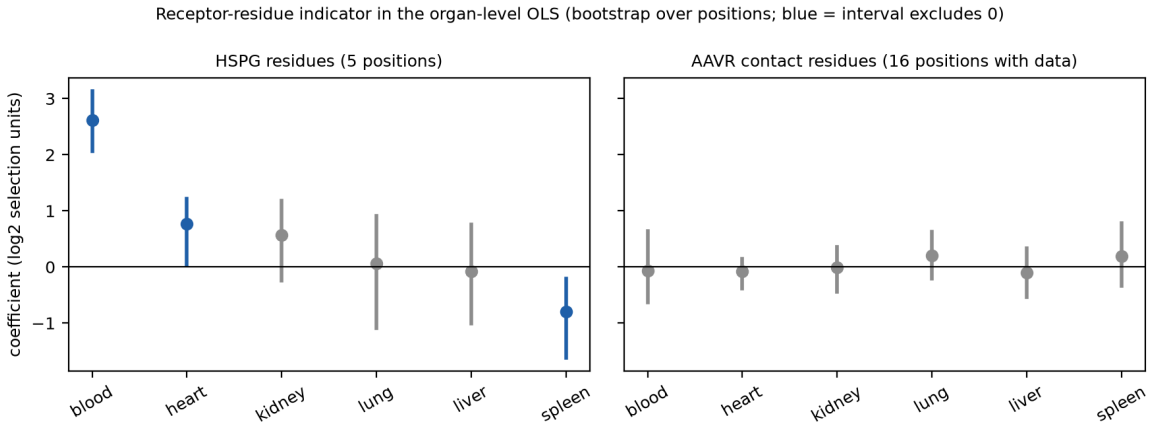


Figure 12: The coefficients in Table 11. Blue: interval excludes zero.

AAVR residues. No tissue shows an AAVR-residue effect beyond the VR effects (Table 11: all six intervals include zero, with point estimates between -0.11 and $+0.20$). At the position level, mean absolute selection is 1.04 vs. 0.98 for region-matched residues ($p = 0.51$); heart (-0.90 vs. -0.70) and liver (-1.65 vs. -1.32) are lower at $p = 0.057$ each, and no tissue reaches 0.05. Four of the six scanned VR-I positions are in the AAVR set, so the VR-I and AAVR terms cannot be separated well: the stable VR-I coefficients in kidney, liver and spleen are no longer stable when the receptor terms are added. Three listed residues have no mutants, and 9 of the 16 positions with data have three or fewer, so the power to detect a per-residue effect is low.

VR-VIII. Table 12 compares the VR-VIII coefficients from the position-level bootstrap with and without the receptor terms. Blood, heart and kidney decrease when the HSPG term is added (blood 1.26 to 1.04, heart 0.29 to 0.23, kidney 0.62 to 0.56), and the heart interval then reaches zero. Lung, liver and spleen are essentially unchanged (lung -0.76 to -0.78, liver -0.51 to -0.49, spleen -0.70 to -0.65) and remain stable. So the negative lung, liver and spleen component of VR-VIII is not carried by R585 and R588. Figure 13 shows the per-position profile (substitutions only). Blood is highest at 585 and 588 (2.7), and several nearby positions are also positive (583: 1.5, 584: 0.8, 586: 1.2, 590: 1.0); the remaining positions range from -1.9 to 0.3. Heart and kidney are positive at 585 and 588 (0.5 to 0.6) and at 583 (0.1 heart, 0.5 kidney) and near or below zero elsewhere. Lung, liver and spleen are negative at nearly every position.

Table 12: VR-VIII coefficient in the organ-level OLS with the bootstrap taken over positions, without and with the HSPG and AAVR indicators (bootstrap mean and 95% interval). Bold: interval excludes zero.

Tissue	Without receptor terms	With receptor terms
Blood	**1.26 [0.85, 1.67]**	**1.04 [0.71, 1.40]**
Heart	**0.29 [0.06, 0.51]**	0.23 [−0.00, 0.45]
Kidney	**0.62 [0.31, 0.93]**	**0.56 [0.26, 0.88]**
Lung	**−0.76 [−1.16, −0.39]**	**−0.78 [−1.18, −0.38]**
Liver	**−0.51 [−0.85, −0.21]**	**−0.49 [−0.83, −0.16]**
Spleen	**−0.70 [−1.01, −0.38]**	**−0.65 [−0.96, −0.32]**

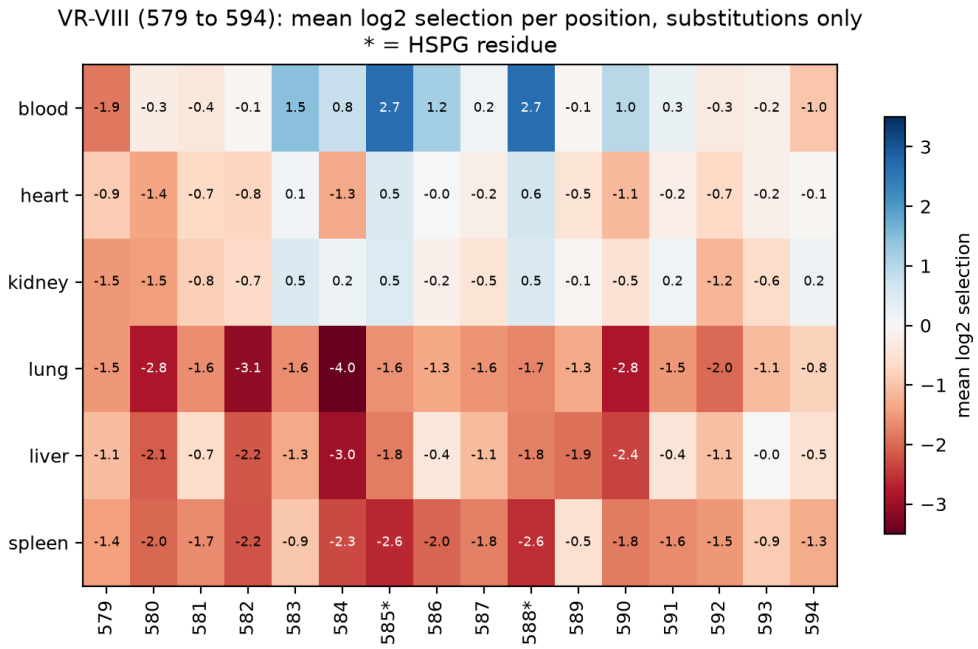


Figure 13: Mean log2 selection per tissue at each VR-VIII position, substitution mutants only. Asterisks mark the HSPG residues R585 and R588.

Summary of this section.

- The HSPG residues differ from their region-mates in blood (higher), heart (higher) and spleen (lower). The direction is what would be expected if less HSPG binding leaves more vector in circulation and less taken up by tissue. That expectation comes from the literature; the data here do not test it.
- In liver the HSPG residues are not distinguishable from region-mates.
- The AAVR residues show no tissue effect beyond the region. AAVR has been reported to be required for AAV infection in cell lines and in mice (Pillay et al., 2016), which would not predict a tissue-specific pattern, but the number of mutants here is too small to tell an absent effect from an undetected one.
- An AAV2 variant with residues 585 to 590 replaced by AAV8 residues (AAV2i8) was reported to have about 40-fold lower liver targeting and to target heart and skeletal muscle (Asokan et al., 2010). The heart direction at R585 and R588 here is the same. The liver result here is not specific to those residues.
- These results are associations at 5 HSPG positions. No test of mechanism was done. Other reported AAV2 co-receptors (integrin, FGFR1, laminin receptor, c-Met) were not analyzed because we did not have verified residue-level footprints for them.

3.9 PCovC and PCA embeddings

The embeddings and cross-validated scores below concern the six-tissue selection profiles, not the mutation features. Figure 14 shows the PCA embedding and PCovC embeddings at three mixing

values, colored by the paper's clusters. Table 13 gives k-nearest neighbor accuracy and silhouette score for two label sets: the paper's clusters (which were themselves derived from PCA and K-means on the same data) and an independent label, the tissue with the largest absolute deviation for each mutant. The embedding was refit inside each training fold. Values are the mean over 100 folds (5 folds, 20 repeats) with the 2.5th to 97.5th percentile of the fold values in brackets; the brackets show the spread of fold scores, not a confidence interval on the mean.

Against the paper's clusters, PCovC at mixing 0.5 gives kNN accuracy 0.957 vs. 0.952 for PCA, and silhouette 0.059 vs. 0.035; the fold ranges overlap. Silhouette rises as mixing decreases (0.175 at mixing 0). Against the independent label, PCovC at mixing 0.5 and PCA are similar (kNN 0.584 vs. 0.583; silhouette −0.180 vs. −0.191). At mixing 0 the independent-label kNN accuracy is 0.683 [0.601, 0.738] vs. 0.583 for PCA. In a 500-draw label-permutation test at mixing 0.5, both label sets score above every permuted draw ($p = 0.002$). Changing the classifier inside PCovC (logistic regression, LDA, linear SVC, ridge) changed kNN accuracy by at most 0.006 for the paper's clusters and 0.019 for the independent label (`classifier_sensitivity.csv`).

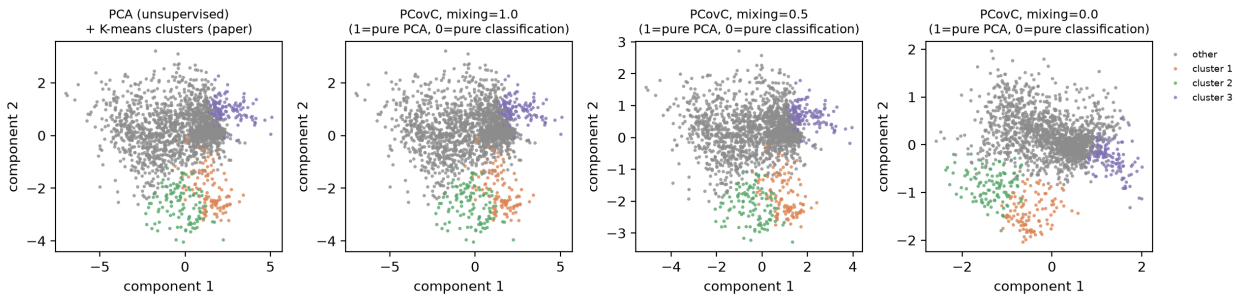


Figure 14: PCA and PCovC (mixing 1.0, 0.5, 0.0) embeddings of the six-tissue log2 selection profiles, colored by the paper's clusters.

Table 13: Nested cross-validated kNN accuracy and silhouette score by method and label set.

Labels	Method	kNN accuracy	Silhouette
Paper clusters	PCA	0.952 [0.935, 0.969]	0.035 [−0.010, 0.078]
	PCovC, mixing 0.00	0.960 [0.941, 0.975]	0.175 [0.113, 0.227]
	PCovC, mixing 0.10	0.964 [0.950, 0.978]	0.129 [0.071, 0.177]
	PCovC, mixing 0.25	0.961 [0.945, 0.976]	0.089 [0.034, 0.137]
	PCovC, mixing 0.50	0.957 [0.943, 0.973]	0.059 [0.009, 0.107]
	PCovC, mixing 0.75	0.954 [0.936, 0.970]	0.044 [−0.003, 0.088]
	PCovC, mixing 0.90	0.952 [0.936, 0.968]	0.038 [−0.007, 0.081]
	PCovC, mixing 1.00	0.952 [0.935, 0.969]	0.035 [−0.010, 0.078]
Tissue of max deviation	PCA	0.583 [0.547, 0.625]	−0.191 [−0.285, −0.092]
	PCovC, mixing 0.00	0.683 [0.601, 0.738]	−0.035 [−0.082, 0.011]
	PCovC, mixing 0.10	0.639 [0.593, 0.678]	−0.086 [−0.146, −0.035]
	PCovC, mixing 0.25	0.568 [0.527, 0.611]	−0.153 [−0.234, −0.074]
	PCovC, mixing 0.50	0.584 [0.549, 0.623]	−0.180 [−0.273, −0.088]
	PCovC, mixing 0.75	0.584 [0.540, 0.628]	−0.188 [−0.282, −0.093]
	PCovC, mixing 0.90	0.583 [0.552, 0.623]	−0.190 [−0.284, −0.092]
	PCovC, mixing 1.00	0.583 [0.547, 0.625]	−0.191 [−0.285, −0.092]

4 Limitations and open items

- Bootstrap resamples in the OLS, p_{xz} and Table 2 results are drawn over mutants. Mutants at the same position share structural context and are not independent, so those intervals are probably too narrow. Resampling by position was done only for the VR coefficients and the receptor model (Section 3.8); there, 30 of 48 VR coefficients are stable rather than 35. The property-delta and p_{xz} results have not been rerun by position.
- The position-level tests treat positions as independent. VR positions are contiguous in sequence.
- The 138 OLS coefficients and 100 p_{xz} weights are not corrected for multiple comparisons. A stable coefficient means only that its bootstrap interval excludes zero.
- The 2-fold axis (n=4), VR-II (n=1), VR-I and VR-IX (n=6 each) and deletions (n=37) are small groups.
- The R^2 comparison is in-sample and between non-nested feature sets.
- The position-level and model-level results for VR-VII have not been reconciled.
- The substitution/insertion sign difference in the centrality relationship has no explanation yet.
- VR boundaries are taken from the authors' code and were not re-derived from the structure. VR-III is excluded.
- The receptor residue sets rest on 5 (HSPG) and 16 (AAVR, with data) positions. The AAVR contact list was read from the article page of Meyer et al. (Table 2). The HSPG results of Opie et al. and Kern et al., and the Asokan et al. and Pillay et al. results cited in Section 3.8, were taken from abstracts or search summaries, not from the full texts.
- Axis membership, RSA and centrality come from a single structure (1LP3) and one contact table. No 3D rendering of any result has been made.

5 Files

Scripts and outputs are in `pcovc_analysis/` and `pcovc_analysis/deep_dive/` under the project directory. Figures used here are copied to `report_figures/`. Tables in this document were generated from the CSVs in `deep_dive/` by a script, not typed by hand.

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